



Research

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Places for reasoning

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In this article, we explore behaviour settings that enable reasoning and the diversity of constraints that not only limit but also make these behaviour settings possible. We focus specifically on reasoning and surveying how behaviour settings allow for the generation of norms of action that are nevertheless differentiated by geographies and sociocultural systems. These geographies and sociocultural systems involve diverse trajectories for reasoning even within similar behaviour settings. We will touch on places for reasoning like Twitter, social movements, traditional knowledge and laboratories set up for experimentation on our reasoning abilities. We will show how these places and the behaviour settings that emerge in them can be studied in terms of the complexity of the interactions between their participants and in terms of enabling constraints.

This article is part of the theme issue ‘People, places, things, and communities: expanding behaviour settings theory in the twenty-first century’.

1. Introduction

Behaviour settings emerge in the interplay among the activities of different agents and the material, social and normative constraints imposed by specific environments and contexts. These settings range from dyadic conversations to scientific meetings, in-person reunions to social media platforms and so on. In this article, we explore several behaviour settings to understand their characteristics, dynamics and, especially, implications for reasoning. We focus on the features and dynamics of each behaviour setting, including the roles of participants, the communication patterns, the social norms and other constraints that make the action patterns typical of each setting emerge. We show that many of these patterns are tied to the particular communities and the specific locations where they take place [1]. For example, dyadic conversations involve interactions between two individuals—in-person, by phone or via text message—which may have distinct dynamics and patterns compared with larger group interactions in, say, scientific meetings or social media platforms. Behaviour settings shape and influence the types of reasoning that may occur in a particular situation, the norms and expectations for reasoning, and the ways in which information is exchanged and processed in those activities. Moreover, attention to behaviour settings allows us to identify commonalities and differences in reasoning patterns. This comparative analysis provides insights into how behaviour settings influence reasoning trajectories and outcomes. It also sheds light on how reasoning activities may be influenced by geographical, political and sociocultural factors. We first provide a succinct presentation of the notion of behaviour settings in the context of ecological psychology. Then, we analyse reasoning and its emergence in different behaviour settings. Lastly, we explore

several varieties of constraints, including *enabling constraints* [2], that allow for reasoning activities and that help frame their study.

2. Behaviour settings

Behaviour settings were introduced by Roger Barker in 1968 [3] to describe self-organized, dynamic eco-behavioural patterns that operate at a scale higher than the individual one. He proposed these patterns are constituted by the collective actions and activities of individuals in conjunction with their material and cultural environment. Through collective actions in particular contexts of natural and sociocultural affordances, groups of individuals generate extra-individual dynamic patterns that constrain individual behaviour and that can be transmitted to future behaviours, including across generations.

Harry Heft characterizes these patterns of actions as quintessentially occurring in and contributing to a *place*: ‘what makes any place distinctive—what gives it significance and meaning—is its affordances for the life activities of an animal’ [1, p. 104]. The repeated emergence of behaviour settings in places allows Heft to emphasize that ‘individuals’ actions and activities degrees of freedom are constrained by previous actions that have generated the settings for action to be possible’ [1, p. 8] and that these actions depend upon other past and current participants in these behavioural settings. The operational dynamics of behaviour settings are best understood by acknowledging that their limitations and opportunities arise from the combined actions of their participants [4]. Consequently, a key reason why individuals generally adhere to the practices of a place is that their engagement in the collective process essentially anchors their individual actions, keeping them within prescribed boundaries. In these collective processes, others often play a guiding role [5, p. 98].

In a place, agents can find significant affordances and can participate in the ongoing self-organization of behaviour settings. From the ecological point of view, perceptual information available in the environment makes affordances available to different kinds of agents. Places are not empty, but are inhabited and transformed through the actions carried out by different types of agents. This implies that there are spatial and temporal limits within the ecological scale. Affordances are not independent of the places they belong to, although they can be transformed in them. In the case of human environments, in which all these changes entail a culture, behaviour settings emerge: ‘[B]ehavioural settings are a notable and distinctive manifestation of our sociocultural nature’ [5, p. 99].

Behaviour settings encompass the physical locations and social environments where people engage in their daily activities, such as classrooms, laboratories, restaurants, social media and so on. Barker’s research revealed that individual actions exhibit consistent predictability when considered within the context of the behaviour setting in which they take place [6]. In other words, everyday behaviours tend to be influenced by and tied to specific behaviour settings. These behaviour settings, in turn, are shaped by the activities carried out within them. Activities shape behavioural settings that embody the normativity that constrains those activities and sometimes generate new ones. Simultaneously, different geographies and sociocultural systems allow the emergence of places that generate particular and situated cognitive strategies. This resonates with the notion of perceptual practice described by Arango [7] in order to characterize the range of ‘intersubjectively constituted, sensory-based practices of interaction with worldly objects’ (p. 32) that operates within human communities, producing constraints on what, when and by whom affordances can be used. The same behaviour setting may provoke different actions for different agents at different times.

Following this characterization of behaviour settings and places, §3 is devoted to seeing how to use behaviour settings as a framework to study reasoning activities. In doing so, we describe ways in which geographies and sociocultural systems contribute to diverse trajectories in reasoning and demonstrate that, even within similar behaviour settings, different reasoning patterns can emerge.

3. Ecological reasoning

One of the activities that can emerge in and constitute a behaviour setting is reasoning [8]. However, it is important to note that, even when authors (e.g. [9]) highlight the importance of the environment to understand human reasoning, typically ‘reasoning is characterized as a set of inference rules susceptible to computational modelling; the only role of the senses is to provide inputs for these basic rules’ [10, p. 117]. Reasoning, that is, is often characterized as disembodied and dis-embedded. An ecological and enactive point of view, however, requires considering interactions in the context of the agent–environment system as constitutive parts of reasoning abilities. Fonseca, for example, notes that ‘an important part of understanding human rationality and heuristic reasoning is to account for how the circumstances of our corporeality, our social and material environment can restrict, shape and alter the use of cognitive strategies’. [10, p. 120]. An ecological approach to reasoning requires an account of the relationship between the activities of agents and the resources of the environment. Behaviour settings for reasoning emerge in this relationship. Furthermore, a static vision of the reasoning process is insufficient: snapshots of reasoning activity do not properly inform about the abilities, norms, rules, values, material culture and so on that constitute the activity. Behaviour settings are extended in space and time: behaviour settings emerge at particular places and through the interactive actions of a number of agents, lasting as long as agents maintain them in that place. By analysing reasoning through the lens of behaviour settings we can (i) explore the generation of reasoning habits and skills, (ii) understand how experience leads to the development of new cognitive strategies, and (iii) account for the parts of the environment—social and material culture and artefacts—that partly constitute reasoning.

A habit is a regulated behavioural pattern formed through past experiences [11,12]. Habits can be learned on purpose but many times, it is the repeated interactions with some material (such as clay) that turn into a particular habit. The

rules of reasoning can be understood as established patterns within the collective activities that shape particular behavioural settings. These reasoning habits position us in relation to the information provided by the environment through affordances, empowering us to make various inferences about the environment and our interactions with it. Habits are ways of using and incorporating the environment [13, p. 15] norms and patterns of thinking or action. According to Dewey:

The influence of habits is decisive because all distinctively human action must be learned, and the very heart, blood and sinews of learning is creation of habitudes. Habits bind us to orderly and established ways of action because they generate ease, skill and interest in things to which we have grown used and because they instigate fear to walk in different ways, and because they leave us incapacitated for the trial of them. Habit does not preclude the use of thought, but it determines the channels within which it operates. Thinking is secreted in the interstices of habits. The sailor, miner, fisherman and farmer think, but their thoughts fall within the framework of accustomed occupations and relationships [14].

Following Dewey's perspective, reasoning can then be understood as a set of habits that create meaning through patterns, aiding us in coordinating future actions and in creating thoughts. Thus, reasoning helps us to discern opportunities the environment presents within particular contexts, guiding us in forming both behavioural and cognitive patterns. These patterns may evolve into new habits that can become shared resources or behavioural norms. Within this general approach, we propose ecological reasoning as the ability, based on habits, to make inferences while constrained by a context of meaning. Humans make meaning by interacting with things, beings, symbols, norms, etc.; humans inhabit interactive environments that are full of sociomaterial utterances. These reasonings are not exclusively abstract or argumentative, they include practical reasonings, ways to handle situations and material circumstances in different arrangements. Following this and looking a bit more closely at material aspects of the environments, the majority of objects, according to Vega [15], possess a technical reality. What this means is that they are not only shaped by technical alterations but also serve as vectors within a constantly evolving space of transformations. These objects guide us in exploring the potential actions the environment offers.

But neither techniques nor artefacts exist apart from the people who apply them or use them. A stone itself has not been made by anyone. It is not an artefact, but it can be used as a means to break or polish another stone, to break a stick or fruit or to kill someone. When someone intentionally uses the stone to transform a specific object, they have produced an artefact with certain purposes and norms. If that artefact enters the habitual use of a group, it will generate a trajectory and that trajectory can be articulated with other artefacts and people, generating further actions, norms and affordances. Artefacts can, therefore, be part of the constitution of new behaviour settings. Those behaviour settings become part of the human experience. Simply put, our lives, from the moment we were born to our last breath, unfold in a world shaped by the cumulative impact of past human endeavours on people and material things. Overlooking this reality leads to the misconception that our experiences occur solely within an individual body and brain. It is essential to recognize that our experiences are not isolated; they have external origins that contribute to their unfolding [16, pp. 39–40].

Artefacts are part of the elements that generate behavioural settings. Thus, material culture must be seen as part of the normative geography of cultural activities, especially since in many cases norms are not explicit but rather implicit and embodied in the agents. In other words, within behaviour settings, cognitive technologies embody cognitive abilities in patterns of activities capable of being inherited (with modifications) from generation to generation, and which usually involve the use of certain artefacts for reasoning. These cognitive technologies constitute skill niches, which promote habits and their reproduction through generations and promote different modalities of types of reasoning and skills.

Experiences that guide behaviour are embedded in contexts or niches that permit the emergence of modalities of interaction that become modes of reasoning. One mode of these reasonings is explicit thoughts, which can be understood as short-lived artefacts produced to deal with the real-time constraints of a behaviour setting. That is, when one says quietly to oneself that 'now would be a good time to advance my queen' in the behaviour setting of a chess match it is both an act of reasoning and the habitual construction of a single-use artefact, designed specifically to deal with the current situation [17].

4. Enabling constraints for reasoning

Although inferential abilities—like many other abilities and skills (e.g. walking, talking)—can be used in a wide variety of situations and, in general, we are able to play and replay inferences at will, not all situations and environments allow for reasoning in the same way. We neither reason equally in different situations nor regard all situations equally good for reasoning. For instance, most people would agree that a properly organized academic conference is a better reasoning environment than Twitter. And we all have 'won' a rational debate in our head hours after the actual debate is over and our interlocutor is not there to answer anymore. This kind of situation illustrates the effect of other agents and conversational dynamics on our inferential abilities—we all 'win' more rational debates in our heads than in actual conversation.

The *ecological* component of ecological reasoning is therefore as important for characterizing reason as whatever intrinsic inferential abilities or skills agents may have. But how can we approach the study of this ecological component? Behaviour settings are handy to answer this question. The emergent dynamics of behaviour settings reflect the elements that make them possible—material environment, affordances, skills, norms, social interactions and so on—by virtue of being constrained by them. Studying the variety of constraints that participate in the emergence of behaviour settings is, therefore, an open route to, first, understanding behavioural settings in general and, second, studying the features of those behaviour settings that we can label as 'places for reasoning'. This is our task in what follows, but a caveat is needed: constraints do not stand for behaviour settings. Constraints are an explanatory tool that can be used to better understand behaviour settings. The material

environment, for instance, is part of a behaviour setting and constrains it in particular ways, but such a constraint does not exhaust its nature. The material environment is more than a constraint, but taking it as a constraint of a higher-order behaviour setting helps us track its relationship with other elements of the setting and, therefore, it is a good route to start understanding the system as a whole.

That said, we will begin with a few words on constraints in general. In the most common conception, constraints are *limitations* that influence and shape the behaviour and dynamics of a given system. As a kind of limitation, constraints often act as barriers that restrict behaviour. They can be an obstacle for conversational patterns or the influence of a visual label indicating a social norm in a particular situation, for instance. However, limitations are not always barriers. Some limitations, physical or otherwise, act as enablers instead. For example, an anatomical constraint such as the bending limitation of the human knee—just one degree of freedom—enables tractable control of bipedal walking. If knees had more degrees of freedom, like the three degrees of freedom we find in ankles or wrists, the control of bipedal walking would be astronomically more difficult if not impossible given our biological resources. Therefore, the bending limitation of the knee is effectively a constraint (among others, of course) that enables a particular functionality in the system. This is the kind of constraint Raja & Anderson [2] have called an *enabling constraint* (e.g. [18]).

Formally, Raja & Anderson [2] define ‘a positive constraint between S and $\{X\}$ that results in strictly functional outcome(s) for S' (p. 214), where S is a given system and $\{X\}$ is a set of entities or processes that influence S . Enabling constraints are ‘positive’ insofar as they result in strictly functional outcomes of S ; namely, enabling constraints result in a system developing new and/or better functions would not be exhibited if those constraints were not in place. Imagine an English Premier League (EPL) football game. In order for it to be such, that is, in order for it to be the kind of behaviour setting we identify as ‘an EPL football game’ many entities and processes must be in place. And many of them will act as a constraint for the game: the proper dimensions of the field, the right number of players, the ball, the clothing, the tactics and so on. However, not all of them are an enabling constraint of an EPL football game. Not all of them make it an EPL football game. For instance, the dimensions of the field are the same for an EPL football game and any other football league in the world. German Bundesliga, Spanish La Liga and Italian Calcio, to name a few, all share the same field dimensions. However, only the EPL football game is played under the authority of the Premier League (as an entity) and The Football Association (FA). These two entities determine what an EPL football game is and what it is not. The normative constraints imposed by the two entities are, therefore, enabling constraints of an EPL football game.

This example highlights two important aspects of the role of (regular and enabling) constraints in behaviour settings. First, it shows the constraints that shape behaviour settings operate at many different, but equally important scales. Constraints can be *physical*. Physical constraints encompass the tangible elements of the environment, including its size and layout, the presence or absence of certain objects or tools, and the availability of resources. For example, a behaviour setting like a laboratory may have specific equipment and apparatus that impose constraints on the types of experiments that can be conducted. Constraints can also be *social*. Social constraints pertain to the norms, rules and expectations that govern behaviour within a setting. They are shaped by social structures, hierarchies, institutions (economical, political and religious) and the established roles and relationships among participants. For instance, in a behaviour setting like a classroom, the presence of a teacher and the established rules of behaviour create social constraints that influence how students interact and reason.

Another kind of constraint—although closely related to the social one—encompasses *cultural* constraints. Cultural constraints refer to the beliefs, values and norms that influence behaviour within a particular setting. Different cultures have distinct norms and expectations regarding reasoning, perception, communication styles and decision-making processes. Cultural constraints shape behaviour settings by providing a framework for what is considered appropriate or acceptable within a specific sociocultural context. We can also find *cognitive* constraints. Cognitive constraints involve the processes, abilities and limitations that impact thinking writ large within a behaviour setting. These constraints can include factors such as attention span, memory capacity, cognitive biases and heuristics. Cognitive constraints can influence how individuals perceive affordances, make inferences when reasoning and arrive at decisions within a specific context. And last but not least, constraints can be *temporal*. Temporal constraints refer to the time-related factors that shape behaviour settings. These constraints include the duration of the setting and temporal expectations or deadlines. Temporal constraints can also impact the pace of interactions within a behaviour setting.

The second relevant aspect regarding constraints and behaviour settings highlighted by the EPL football game has to do with the way enabling constraints relate to the emergence of a particular setting. The example shows that at least one of the enabling constraints that allows for the emerging of the setting known as ‘an EPL game’ is sociocultural. The relation between constraints at different scales is not hierarchical and an enabling constraint may appear at any of the possible scales. At the end of the day, it is the network of constraints and other elements, including the interactions between different agents, that makes a behaviour setting emerge. In this context, the study and analysis of the diversity of constraints within behaviour crucial to understanding them. More particularly, it is crucial to understand the different characters of different behaviour settings and what makes them the kind of setting they are, enabling constraints are the tool.

In the concrete case of reasoning, paying attention to the constraints that lead to the emergence and evolution of behaviour settings that involve reasoning may be an open avenue to understanding reasoning itself but also its conditions of possibility. In other words, a constraints-based analysis may tell us whether a behaviour setting is a place for reasoning and, if so, what enables the emergence of such a place. By examining the interplay between constraints and reasoning, we can gain insights into how the specific features and dynamics of behaviour settings influence the generation and expression of reasoning processes and behaviour outcomes.

The usefulness of the notions of behaviour settings and (enabling) constraints to study reasoning and places for reasoning is twofold. As in the EPL game example, the analysis of the behaviour setting and its constraints in the context of reasoning

may be *narrative*. By this, we mean the description of the behaviour setting and its relevant constraints in plain language, so to speak. In the case of the EPL game, we have mentioned elements like the size of the field, the number of players, the kind of ball, the clothing, the normative environment, etc. A list that allowed us to explore which constraints genuinely enable such behaviour settings. This kind of list may of course be as exhaustive as the aims of the researcher require. In the case of reasoning, we can apply the same strategy to different places for reasoning. For instance, to be a place of reasoning, a debate in the context of a session in a conference must meet requirements of many different kinds: a *physical* environment that allows for interaction—proper layout and illumination for the participants to see each other, microphones and speakers if needed, etc.—a set of *social* norms specific of the kind of reasoning expected—maybe a moderator or a chair of the debate, time enough to perform the needed inference (this constraint would be as social as *temporal*), turns to speak, etc.—a background of *cultural* norms that harnesses all interactions—paying attention to other people, being in silence when others talk, etc.—the proper *skills* to participate in the conversation—speaking the language, knowing the topic, etc.—and so on.

A different place of reasoning might share some of these elements. For example, if we take a Twitter debate to be a place for reasoning, we will find the physical, social, cultural and ability constraints that make it this kind of behaviour setting. However, many of these constraints will not be shared with a debate at a session of a conference. Physically, Twitter needs some form of working electronic device, not the layout of a room. There is a need for moderators, but their task is likely different from the ones in the debate. The attention paid to the interlocutor may fluctuate and silence does not seem to be a requirement. And the skills and temporal constraints are also diffused. There are, however, a few aspects that typically must remain constant for reasoning to truly emerge: the necessity of minimal competence in inferential activities and the necessity of some form of interactive engagement with the interlocutor. We may identify these two particular elements as enabling constraints for a place for reasoning and claim they are unavoidable for any behaviour setting to be regarded as such a place. Whether this assessment about the enabling constraints on reasoning is right or not is a matter of empirical investigation, but the analysis highlights two benefits of exploring places of reasoning in terms of behaviour settings. The first benefit has to do with the systematic description of the behaviour setting at multiple scales. Such a systematic description allows for the careful identification of constraints and whether they are enabling or not. And the second benefit is that after the systematic description of two behaviour settings as places for reasoning (or not), they can be compared. For instance, we could wonder whether a session at a conference is a better place for reasoning than Twitter in the context of a debate. By evaluating how each behaviour setting related to the enabling constraints of reasoning and how the rest of the constraints support or go against the enabling ones, we would be in principle able to assess which behaviour setting is a better place for reasoning.

Importantly, this kind of analysis does not need to be just narrative, but it can also be implemented in terms of modelling. In principle, it is possible to characterize behaviour settings using the usual formal methods of ecological psychology and include the effect of different kinds of constraints in them (e.g. [19]). Typical formal methods used in ecological psychology are dynamical systems theory and complexity analyses (e.g. [20,21]). In the case of dynamical systems theory, the change over time of a behaviour setting (i.e. its dynamics) can be characterized as a function of some of its variables when constrained for some parameters. For example, we can characterize a group of people walking while following a leader—a human flock—in terms of their global or individual steering dynamics (e.g. [22,23]). To do so, we can find the equation that describes their changes in steering given constraints like their proximity to other members of the flock (a physical constraint) or the presence of a leader (a social constraint). If we take following a leader around in a particular environment to be a behaviour setting, then we would have a formal model of the setting that includes both physical and social constraints. We could even simulate the model including, subtracting and manipulating constraints in order to understand how they shape the emergence of the particular behaviour setting and whether one or some of them is an enabling constraint.

As an example, we focus here on *interpersonal synergies* as enabling constraints in two different behaviour settings. A synergy is a collection of components that act as a temporary self-organizing unit, and in which the components exhibit behaviour that is constrained by being parts of the synergy. An interpersonal synergy is a synergy in which the components are individual human agents. Humans routinely form interpersonal synergies and doing so is beneficial. Interpersonal synergies are our basic bodily engagements with one another, the glue that makes possible more complex human interactions like having a conversation, playing a game or making music. Interpersonal synergies, that is, act as enabling constraints in these behaviour settings. In 1999, Chartrand & [24] described the ‘chameleon effect’, which is the tendency for interacting humans to imitate one another’s facial expressions, intonations and gestures while engaged in conversation. Starting a few years later, cognitive scientists studying embodied social interaction focused on unintentional interpersonal synchrony. Shockley *et al.* [25] showed that people engaged in conversation and joint problem solving unintentionally synchronized their postural sway and their gazes; Richardson *et al.* [26] showed that people who sat in rocking chairs while engaging in conversation and joint problem solving unintentionally synchronized their rocking; Ramsmeier & Tsacher [27] showed that patients and therapists unintentionally synchronized their movements and that this synchrony improved patient outcomes. Paxton & Dale [28] showed that affiliative conversations are accompanied by more bodily synchrony than arguments, even though both still involve close coordination; arguing perturbs, but does not disrupt interpersonal synergies. More generally, to echo Fusaroli & Tylén [29], all dialogues are interpersonal synergies. Interpersonal synergies are enabling constraints in dyadic conversation behaviour settings.

A second example: Abney *et al.* [30,31] a two-person building game in the laboratory. They had pairs of participants collaboratively build towers from marshmallows and uncooked spaghetti on a tabletop. One participant could touch only the spaghetti and the other could touch only the marshmallows. They were told to build the tallest tower they could in 15 minutes. This is admittedly a strange task, but it is open-ended and relatively unconstrained; it also requires cooperative reasoning and relatively natural conversation from the participants. They found that vocal complexity matching, a measure of the strength of interpersonal synergy, predicts success at the task. Pairs who formed stronger interpersonal synergies built taller towers. Along

with the rules of the game and various physical features of the laboratory, interpersonal synergies are enabling constraints in this place for reasoning.

The same principles can be applied to other places for reasoning. We could, for instance, capture the dynamics of a debate in the context of a round table by measuring the changes in a meaningful variable like the level of agreement or disagreement through the interaction. And we could check how parameters that reflect constraints like the necessity of minimal competence on inferential activities and the necessity of some form of interactive engagement with the interlocutor influence such change. By doing so, we would characterize a place for reasoning in terms of (enabling) constraints of different scales and types. From this point, we could compare different places or reasoning or evaluate different properties of some of them—e.g. which constraint(s) are responsible for the increasing (or decreasing) of polarization in Twitter debates. Therefore, characterizing places for reasoning in terms of behaviour settings and constraints opens a wide range of opportunities to understand reasoning activities.

5. Conclusions

The ecological facet of reasoning is as crucial as the intrinsic inferential skills possessed by agents. Exploring this ecological dimension leads us to behaviour practical avenues encapsulating emergent dynamics influenced by elements such as the material environment, affordances, skills, norms and social interactions. By dissecting the diverse constraints shaping these settings, we pave the way for a universal understanding, specifically examining the characteristics designating them as ‘places for reasoning’. This exploration into the interplay between constraints and reasoning provides valuable insights into how the features and dynamics of behaviour settings influence the generation and expression of reasoning processes and subsequent behavioural outcomes. Also, discerning these constraints may lead to good experimental outcomes to analyse multiple expressions of reasoning and behavioural outcomes.

Understanding reasoning in ecological terms has potential benefits. In future work, we hope to analyse the different constraints under which behaviour settings for reasoning form in particular communities. A comparative study of the constraints that enable behaviour settings for reasoning can provide more possibilities for understanding how conflicts and epistemic injustices do or do not emerge in those settings.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

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All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

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